

KAUHAJOKI KUJA-KOKKO, Finland

WMO: 027690

Lat: 62.4133N

Lon: 22.1836E

Elev: 305

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.1	(c) -6.2	(d) -16.5	(e) 2.3	(f) -12.0	(g) -10.3	(h) 3.2	(i) -5.8	(j) 23.2	(k) 31.3	(l) 20.9	(m) 31.5	(n) 2.1	(o) 160	(p) 0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.1	(c) 79.1	(d) 64.9	(e) 75.6	(f) 63.0	(g) 72.2	(h) 61.6	(i) 67.7	(j) 75.0	(k) 65.3	(l) 72.3	(m) 63.2	(n) 70.2	(o) 7.6	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 65.1	(b) 94.0	(c) 71.1	(d) 62.5	(e) 85.5	(f) 68.6	(g) 60.3	(h) 79.2	(i) 66.3	(j) 32.2	(k) 75.3	(l) 30.4	(m) 72.1	(n) 28.8	(o) 70.3	(p) 74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	16.6	14.7	-16.6	82.5	7.2	5.1	-21.8	86.1	-26.0	89.1	-30.0	92.0	-35.2	95.7
			-15.7	69.4	8.1	3.7	-21.5	72.1	-26.3	74.2	-30.9	76.2	-36.8	78.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	18.7	21.3	26.8	37.1	48.4	54.9	60.5	57.4	50.3	38.8	32.9	26.1		
	DBStd	16.27	11.95	12.07	9.20	6.30	7.85	6.14	5.84	5.62	6.03	6.73	7.93	10.66		
	HDD50	4702	970	803	719	389	124	20	1	6	68	349	514	739		
	HDD65	9337	1435	1223	1184	837	518	310	165	245	441	811	964	1204		
	CDD50	885	0	0	0	2	73	167	327	236	77	3	0	0		
	CDD65	44	0	0	0	0	2	6	26	10	0	0	0	0		
	CDH74	471	0	0	0	0	40	81	281	69	0	0	0	0		
	CDH80	72	0	0	0	0	1	7	53	11	0	0	0	0		
	WSAvg	7.0	6.6	7.3	7.4	7.0	7.4	7.2	6.4	6.7	6.4	6.9	7.2	7.5		
	PrecAvg	25.7	1.9	1.4	1.3	1.1	1.7	2.5	3.1	3.0	2.4	2.7	2.3	2.3		
	PrecMax	32.3	4.1	2.9	2.2	2.2	3.1	5.4	5.6	5.4	5.3	6.0	5.0	4.1		
(15) Precipitation	PrecMin	20.2	0.4	0.0	0.3	0.2	0.4	0.7	0.2	0.7	0.7	0.6	0.4	0.7		
	PrecStd	3.1	0.9	0.7	0.4	0.6	0.8	1.0	1.3	1.2	1.3	1.3	0.9	1.0		
	DB	40.0	42.2	48.6	66.4	78.6	80.8	85.4	81.2	69.1	56.6	50.4	44.4			
	MCWB	38.7	38.2	39.7	50.4	59.5	64.1	68.3	68.9	61.2	53.0	48.8	42.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	37.3	39.1	43.4	57.7	73.8	76.1	80.5	75.1	65.3	53.1	46.8	41.0			
	2% MCWB	35.7	36.5	38.3	46.4	58.3	62.0	66.3	64.4	58.4	50.6	45.5	39.6			
	5% DB	35.0	36.5	40.3	53.1	69.4	71.9	76.8	71.5	62.6	51.2	44.2	38.6			
	5% MCWB	33.9	35.1	36.4	44.1	56.0	60.0	64.6	61.8	56.7	49.0	43.1	37.4			
	10% DB	33.4	34.6	38.0	48.8	64.4	67.7	73.3	68.1	60.2	49.0	42.3	36.9			
	10% MCWB	32.6	33.6	35.0	41.9	53.3	58.0	63.7	60.3	55.8	47.1	41.4	36.0			
	0.4% WB	39.3	38.6	41.3	52.3	62.6	67.5	71.0	70.4	62.2	53.8	48.9	43.2			
	0.4% MCDB	40.0	40.7	45.6	61.1	73.9	74.9	80.1	77.1	67.7	55.6	50.2	44.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.2	36.7	38.6	48.0	60.2	64.4	68.5	66.4	59.7	51.6	45.7	39.9			
	2% MCDB	37.1	38.6	42.2	57.7	70.4	73.6	76.1	72.1	63.3	52.7	46.8	40.9			
	5% WB	34.1	35.3	37.0	45.2	57.4	61.5	66.4	63.6	58.1	49.5	43.4	37.6			
	5% MCDB	34.8	36.6	39.8	51.5	67.3	69.5	73.5	69.4	61.4	51.0	44.1	38.3			
	10% WB	32.8	33.8	35.3	42.7	54.0	59.2	64.5	61.7	56.4	47.2	41.6	36.0			
	10% MCDB	33.5	34.6	37.8	48.5	63.7	66.5	71.8	66.6	59.5	48.7	42.3	36.7			
(35) Mean Daily Temperature Range	MDBR	11.2	11.7	14.6	16.0	19.2	17.5	18.1	17.2	14.8	11.0	7.1	8.8			
	5% DB	7.5	9.4	14.0	25.8	27.7	25.7	24.2	22.4	18.7	13.1	8.3	7.8			
	5% WB	7.1	8.0	9.8	15.5	13.7	13.4	11.7	12.3	12.5	10.7	7.8	7.6			
	MCDBR	7.2	9.6	12.2	22.2	24.6	21.1	20.5	17.5	15.7	11.7	8.2	8.0			
	MCWBR	7.0	8.4	9.1	14.3	12.9	11.9	11.3	10.3	11.2	10.4	7.8	7.9			
(40) Clear-Sky Solar Irradiance	taub	0.245	0.297	0.320	0.360	0.356	0.352	0.375	0.361	0.337	0.319	0.265	0.352			
	taud	2.080	2.193	2.282	2.333	2.421	2.455	2.412	2.442	2.501	2.388	2.156	2.565			
	Ebn at Noon	132	205	248	262	274	277	266	260	245	200	122	79			
	Edh at Noon	11	21	27	32	32	31	32	29	23	17	10	6			
(44) All-Sky Solar Radiation	RadAvg	58	236	715	1182	1623	1765	1645	1263	750	300	76	30			
	RadStd	9	51	78	140	165	138	153	125	81	60	14	3			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	-234	N/A	N/A	N/A

Nomenclature: See separate page